Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 24: Approved Solutions

Proof by Cases

Covers Chartrand & Zhang, Section 3.4

Status of this set

Ungraded practice. Solutions released with it. Write every proof in complete sentences, and label every case.

Part I: How to Practice This Material

The template

1. **Say what the cases are** and why they cover everything: "We consider two cases, according to whether n is even or odd."
2. **Label each case** and prove it as a self-contained direct proof.
3. **Close** by noting that the cases are exhaustive, so the result holds throughout.

When to reach for cases

When the hypothesis is too weak to give you an equation. "Let n be an integer" hands you nothing; "let n be an even integer" hands you $n = 2a$.
Splitting manufactures the missing information.

Three mistakes to expect

1. **Cases that miss something.** They must cover the whole domain.
2. **Reusing a letter inside one case.** $m = 2a$ and $n = 2a$ asserts $m = n$.
3. **"Without loss of generality" on cases that are not symmetric.** Only use it when swapping names reproduces the argument exactly.

Part II: Practice Problems

Section A — One variable

1. Prove that $n^2 + n + 6$ is even for every integer n.

Approved solution

Proof. Let n be an integer. We consider two cases.
Case 1: n is even. Then $n = 2a$ for some integer a, and

$$n^2 + n + 6 = 4a^2 + 2a + 6 = 2(2a^2 + a + 3).$$

Since $2a^2 + a + 3$ is an integer, the expression is even.
Case 2: n is odd. Then $n = 2b + 1$ for some integer b, and

$$n^2 + n + 6 = (4b^2 + 4b + 1) + (2b + 1) + 6 = 4b^2 + 6b + 8 = 2(2b^2 + 3b + 4).$$

Since $2b^2 + 3b + 4$ is an integer, the expression is even.
Every integer is even or odd, so the result holds for all n. ■

Shorter route: $n^2 + n = n(n + 1)$ is a product of consecutive integers, hence even, and adding 6 keeps it even. Both proofs are fine; noticing the factorisation is faster.

2. Prove that $n^2 + 5n + 2$ is even for every integer n.

Approved solution

Proof. Let n be an integer.
Case 1: $n = 2a$. Then $n^2 + 5n + 2 = 4a^2 + 10a + 2 = 2(2a^2 + 5a + 1)$, which is even since $2a^2 + 5a + 1$ is an integer.
Case 2: $n = 2b + 1$. Then

$$n^2 + 5n + 2 = (4b^2 + 4b + 1) + (10b + 5) + 2 = 4b^2 + 14b + 8 = 2(2b^2 + 7b + 4),$$

which is even.
The cases are exhaustive, so the result holds. ■

Watch the constants in Case 2: $1 + 5 + 2 = 8$, which is even. If they had summed to something odd the claim would be false — so that arithmetic is worth checking before you start.

Section B — Two variables

3. Let m and n be integers. Prove that $m + n$ is even if and only if m and n are of the same parity.

Approved solution

Two implications, each with two cases.
($\Leftarrow$) Direct. Assume m and n are of the same parity.
Case 1: both even. Then $m = 2a$, $n = 2b$ for integers a, b, and $m + n = 2(a + b)$, which is even.
Case 2: both odd. Then $m = 2a + 1$, $n = 2b + 1$, and $m + n = 2a + 2b + 2 = 2(a + b + 1)$, which is even.
($\Rightarrow$) Contrapositive. We show: if m and n are of opposite parity, then $m + n$ is odd.
Case 1: m even, n odd. Then $m = 2a$, $n = 2b + 1$, and $m + n = 2(a + b) + 1$, odd.
Case 2: m odd, n even. Then $m = 2a + 1$, $n = 2b$, and $m + n = 2(a + b) + 1$, odd.
Both directions hold. ■

The two cases in the second half are symmetric, so "without loss of generality assume m is even" would be legitimate here. Writing both is also fine, and safer while you are learning to judge symmetry.

4. Let m and n be integers. Prove that $3n+1$ and $5n+2$ are of opposite parity.

Approved solution

Proof. Let n be an integer.
Case 1: n is even, say $n = 2a$. Then $3n+1 = 6a+1 = 2(3a)+1$ is odd, and $5n+2 = 10a+2 = 2(5a+1)$ is even. Opposite parity.
Case 2: n is odd, say $n = 2b+1$. Then $3n+1 = 6b+4 = 2(3b+2)$ is even, and $5n+2 = 10b+7 = 2(5b+3)+1$ is odd. Opposite parity.
In both cases the two expressions have opposite parity. ■

Each case requires two computations, one per expression. Students who compute only one have shown half of each case.

Section C — Cases that are not parity

5. Prove that $|x| \geq x$ for every real number x.

Approved solution

Proof. Let x be a real number. We consider two cases.
Case 1: $x \geq 0$. Then $|x| = x$ by definition, so $|x| \geq x$.
Case 2: $x < 0$. Then $|x| = -x$, which is positive, while x is negative. So $|x| > 0 > x$, and in particular $|x| \geq x$.
Every real number satisfies $x \geq 0$ or $x < 0$, so the result holds. ■

The cases came from the definition of $|x|$, which is itself defined by cases. When a definition splits, the proof usually splits the same way.

6. Let A, B, C be sets. Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, using cases rather than a membership table.

Approved solution

Half one. Let $x \in A \cap (B \cup C)$. Then $x \in A$, and $x \in B$ or $x \in C$.
Case 1: $x \in B$. With $x \in A$ this gives $x \in A \cap B$.
Case 2: $x \in C$. With $x \in A$ this gives $x \in A \cap C$.
Either way $x \in (A \cap B) \cup (A \cap C)$.
Half two. Let $y \in (A \cap B) \cup (A \cap C)$, so $y \in A \cap B$ or $y \in A \cap C$. In both cases $y \in A$.
Case 1: $y \in A \cap B$, so $y \in B$, hence $y \in B \cup C$.
Case 2: $y \in A \cap C$, so $y \in C$, hence $y \in B \cup C$.
Either way $y \in A$ and $y \in B \cup C$, so $y \in A \cap (B \cup C)$.
The two inclusions give equality. ■

The cases came from a disjunction in the hypothesis. That is the most common source of a case split outside parity, and you used it on the midterm without necessarily naming it.

Section D — Writing

7. In five to eight complete sentences, explain why the cases in a proof by cases must cover the entire domain, and what goes wrong if they do not. Give a specific example of a case split that fails to cover its domain.

Approved solution

Model answer. A proof by cases establishes a universal statement by proving it separately on pieces of the domain, so the conclusion only reaches elements that some case actually covers. If the cases omit part of the domain, the statement is simply unproved for those elements, and since the claim was universal the whole proof fails — one uncovered element could be a counterexample. This is why the standard integer split into even and odd works: every integer is exactly one of the two, so the pieces form a partition and nothing is missed. A failing example would be proving a claim about all real numbers by considering the cases $x > 0$ and $x < 0$, which leaves out $x = 0$ entirely. That gap is easy to overlook and is exactly where such claims tend to break. The remedy is to state the cases explicitly and check that every element of the domain satisfies at least one.

Full credit requires a specific gap, not just "you might miss something." The $x = 0$ omission is the classic one; "positive and negative integers" omitting zero also works.

Section E — Challenge (optional)

8. Prove that for every integer n, exactly one of n, $n+1$, $n+2$ is divisible by 3.

Approved solution

Proof. Let n be an integer. On division by 3, n leaves a remainder of 0, 1, or 2, and exactly one of these. We consider the three cases.
Case 1: $n = 3k$. Then n is divisible by 3. Also $n+1 = 3k+1$ and $n+2 = 3k+2$ leave remainders 1 and 2, so neither is divisible by 3.
Case 2: $n = 3k+1$. Then $n+2 = 3k+3 = 3(k+1)$ is divisible by 3, while n and $n+1 = 3k+2$ leave remainders 1 and 2.
Case 3: $n = 3k+2$. Then $n+1 = 3k+3 = 3(k+1)$ is divisible by 3, while n and $n+2 = 3k+4 = 3(k+1)+1$ leave remainders 2 and 1.
In every case exactly one of the three is divisible by 3. ■

Three cases, not two. The even/odd split is the most common but not the only one — and here it would be useless. **Choose the split that matches the claim**: a claim about divisibility by 3 wants remainders modulo 3.
Note also that "exactly one" required checking, in each case, both that one *is* divisible and that the other two are *not*.

Instructor notes

- **Problem 7 is the conceptual anchor.** The $x = 0$ gap is the classic failure and students find it memorable.
- **Problem 8** breaks the assumption that cases always means two. A claim about divisibility by 3 wants three cases — worth board time.

- **Problem 6** shows a case split arising from a disjunctive hypothesis, which is the same move students used on the midterm's set proof. Naming it retroactively helps.
- **Problems 1 and 2** both have shorter factorisation routes. Mention them, but do not let students conclude cases are second-best — cases always work.
- **Problem 4** needs two computations per case. Half-done cases are common.
- Reserved for Homework 10: Section 3.4 even-numbered exercises. Odd ones have printed answers and are self-check only.